

Karthik Raja KANDAVEL

📍 Lyon, France ✉ karthikrajakandavel@gmail.com in karthik-raja-k 🌐 karthik-raja17 📁 Portfolio

AI Engineer who builds idempotent BigQuery pipelines on GCP using SQL, Python, and data modeling to deliver scalable ETL/ELT with 99.9% reliability, validation, and dashboards.

Education

EURECOM (Institut Mines-Télécom - IMT), France Master of Science in Computer Science (Specialisation in Data Science)	Sept 2024 - Sept 2026
Sri Sairam Institute of Technology, India B.E in Computer Science and Engineering	Jun 2018 - Jun 2022

Technical Skills

- **AI/NLP & Agentic Systems:** LLMs, Agentic RAG, Knowledge Graphs, Hybrid Retrieval (BGE-M3/BM25)
- **Programming:** Python (FastAPI), SQL (BigQuery, PostgreSQL), Bash/Unix, Git, REST/Webhook APIs
- **Integration & MLOps:** Docker, CI/CD, REST/Webhook APIs, GCP (Cloud Run, Pub/Sub), Azure (AZ-900)
- **ML & Optimization:** PyTorch, XGBoost, Scikit-learn, MPC (MILP), Digital Twin Simulation, SHAP, CVXPY
- **BI & Communication:** Power BI, Tableau, Looker, Technical Docs, Stakeholder Presentations

Professional Experience

Data Science & ML Engineering Intern (R&D to Production) Faradae	Lyon, France Mar 2026 - Present
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- **ETL/ELT pipelines at scale:** Architected GCP middleware (Pub/Sub, Cloud Run, BigQuery) ingesting live MQTT & billing APIs into BigQuery warehouse every minute; engineered re-runnable, idempotent pipelines with merge/upsert and fallback logic for **99.9% reliability** across 3 sites.
- **Data modeling & warehouse maintenance:** Modeled raw telemetry into clean fact/dimension tables at correct granularity; managed full vs. incremental loads for continuous 1-min scheduling without duplication.
- **Business impact & validation:** Delivered Python/SQL financial pipeline (spot vs. fixed) achieving **1.33x import cost efficiency (25% monthly reduction during the pilot)** at a pilot site; built validation tests + drift monitoring (PSI/KS) logged to BigQuery with alerts.
- **Python automation & dashboards:** Containerized XGBoost microservice (Docker, CI/CD) with monthly retraining vs. champion and Looker monitoring, cutting MAE by **45.1%** vs. baseline.

Recommendation from CEO Thomas Lawson and CFO Michel Lerendu: [LinkedIn Post](#) 📄

Associate Business Analyst (Data Focus) Infosys LTD	Chennai, India Aug 2022 - Aug 2024
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- **Engineered automated data workflows** using SQL and Spotfire, eliminating manual reporting overhead and accelerating data-driven decision-making for client stakeholders.
- **Resolved critical data integrity failures** by diagnosing sensor-level faults and implementing a root-cause fix, restoring accuracy to month-long historical datasets.
- **Led technical reviews and mentored 3 junior developers**, collaborating in an Agile/Jira environment with daily stand-ups and sprint planning.

AI & Data Science Projects

Production-Grade Legal RAG (Personal → Company Adoption) Link - 🌐 📄 <i>ChromaDB, BGE-M3, BM25, Vertex AI, Groq Llama-3.1, GCP (Cloud Run, Pub/Sub, Firestore)</i>	Feb 2026 - Present
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- **From personal prototype to company asset:** Demoed self-correcting agentic RAG to Faradae leadership, securing GCP credits and resources; designed decoupled parser/indexer with async Pub/Sub, scalable to 1,800+ pages.
- **Production-grade indexing pipeline:** Parallelised Dedoc OCR + spaCy French NER into 512-token hierarchical chunks (breadcrumb preservation), improving embedding MRR with idempotent BigQuery logging.
- **Hybrid retrieval & grounded generation:** Integrated query rewriting and cross-encoder reranking (RRF K=60 over 60 candidates), achieving **95.6% citation accuracy** and **100% absent-clause refusal** with 99.9% availability.

AI-Assisted Message Routing in Pub/Sub Systems Link - 🌐 📄 <i>Python, MQTT, Groq AI (Llama3), ThreadPoolExecutor, Dynamic Rule Engine</i>	Mar 2025 - Jun 2025
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- **Architected a hybrid MQTT broker** integrating an LLM (Llama3-70B) for real-time contextual analysis with a dynamic rule engine, adding an LRU cache and failover mechanism ensuring seamless rule-only operation during LLM connectivity loss.
- **Achieved 87% anomaly detection accuracy**, a 25% relative improvement over legacy rule-only systems, validated via a comparative benchmark harness measuring latency and routing accuracy.

Certifications & Relevant Coursework & Languages

- **Microsoft Azure Fundamentals (AZ-900)** Feb 2021
- **EURECOM Technical Mastery:** Achieved 18.5/20 in Software Development Methodologies and 17.5/20 in Algorithmic Machine Learning; training in Optimization Theory, Deep Learning, Distributed Systems (Cloud).
- **English:** C2 (Proficient / Bilingual) • **French:** B1 (Intermediate / Progressing) • **Tamil:** Native